Database Design Principles

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Introduction to Database Design

Database design is the process of producing a detailed data model for a database system. A well-designed database minimizes redundancy, ensures data integrity, and supports efficient querying. Poor design leads to update anomalies, wasted storage, and poor query performance. The design process starts with requirements gathering, then conceptual modeling, logical design, and physical implementation.

Entity-Relationship (ER) Modeling

ER diagrams visually represent entities (objects), attributes (properties), and relationships between them. Entity types: Weak entities depend on strong entities (e.g., OrderItem depends on Order). Relationship cardinalities: One-to-One (1:1), One-to-Many (1:N), Many-to-Many (M:N). A many-to-many relationship is implemented with a junction/bridge table. Attributes can be simple, composite, multi-valued, or derived.

Normalization

Normalization is the process of organizing data to reduce redundancy and improve integrity. 1NF (First Normal Form): Eliminate repeating groups; each column holds atomic (indivisible) values; each row is unique. 2NF: Must be in 1NF; eliminate partial dependencies (non-key attributes must depend on the WHOLE primary key, not part of it). 3NF: Must be in 2NF; eliminate transitive dependencies (non-key attributes must depend only on the primary key). BCNF: Stricter than 3NF; every determinant must be a candidate key.

Primary, Foreign Keys, and Constraints

Primary Key: Uniquely identifies each row in a table. Cannot be NULL. Should be stable (never change). Surrogate keys (AUTO_INCREMENT integers or UUIDs) are preferred over natural keys. Foreign Key: A column that references the primary key of another table, enforcing referential integrity. Cascade options: ON DELETE CASCADE removes dependent rows. ON UPDATE CASCADE propagates key changes. CHECK constraints validate column values. UNIQUE constraints ensure no duplicate values.

Indexing Strategy

B-tree indexes work well for equality, range queries, and ORDER BY operations. They are the default index type in most databases. Hash indexes are ideal for equality lookups but do not support range queries. Covering indexes include all columns needed by a query, allowing the database to answer the query entirely from the index without accessing the table (index-only scan). Composite indexes on (col_a, col_b) support queries filtering on col_a alone or both, but NOT on col_b alone. Avoid over-indexing: indexes speed up reads but slow down INSERT, UPDATE, and DELETE.

ACID Properties

ACID guarantees database transaction reliability. Atomicity: A transaction either completes entirely or has no effect at all. If a transfer of funds fails halfway, the entire transaction is rolled back. Consistency: A transaction brings the database from one valid state to another valid state, respecting all constraints and rules. Isolation: Concurrent transactions execute as if they were serial. Isolation levels: Read Uncommitted, Read Committed, Repeatable Read, Serializable. Durability: Once a transaction is committed, its changes are permanent even if the system crashes, guaranteed through write-ahead logging (WAL).